

Tests and Homeworks

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Economic Complexity, Class 2 – Sept 23 2026

Social Data Science, Corvinus University of Budapest

Quiz 1 – Sept 23

1. What are the assumptions behind the product space? Describe at least 2. (2 points)
2. What are the empirical observations made with the use of product space regarding diversification? (1 point)

HW1 for Sept 28

Interesting problem and data.

1. Identify a topic that is connected to the course and interests you. This can be important for a company, city, an industry, a country, for specific groups of people or the global humanity (eg. automation, inequality, international competition, climate change, populism etc.).
2. Search for journal articles that address this topic with interesting data. Choose one that you would like to show to the class.
3. Prepare 1 pptx slide that summarizes the paper. The presentation cannot be longer than 1 slide. Best slides typically include figures that can illustrate the story.
4. Be able to present the slide and talk about the research problem, question, the data, methods, and findings. Save the file: "givenname_familyname_EC2026_HW1.pptx".
5. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
6. Homeworks covering 1-2-3 can get 2 points. Presentation gives an extra 1 point.

Reading for – Sept 30

Hausmann and Hidalgo 2009 PNAS.

Extra homework for Sept 30

Illustrate the dynamics of complexity ranking in trade, technology and research.

1. Go to the country Complexity Rankings on oec.world and download the country data.
2. Investigate ranking dynamics in all dimensions across 2000-2010-2020. Compare correlations across time but within dimensions. Check cross-correlations across dimensions.
3. /Plus: Merge data with Economic Growth forecast tables <https://oec.world/en/resources/methods/eci-growth-predictions>. Calculate the coefficient of ECI indices using the Solow model of economic growth.
4. Prepare visualizations and discuss the results. Collect figures and text to a pdf (max 2 pages) and save it "givenname_familyname_EC2026_HW1_extra.pdf".
5. Include code and prompts in a zip file "givenname_familyname_EC2026_HW1_documentation.zip".
6. Send it to the instructor and TA by email. Deadline: September 29, Tuesday, 12noon. Homework sent after the deadline are not considered.
7. Extra homeworks covering points 1-2-4-5 can get 2 points. Point 3 gets plus 1 point.